The Method of Moments for VHF/UHF Wire Antennas: From the Pocklington

Electric-Field Integral Equation to a Dual-Band 144–432 MHz Design

An essay for readers working in mathematical physics, functional and numerical analysis, and classical electromagnetism.

Abstract

We give a self-contained derivation of the thin-wire electric-field integral equation (EFIE) for a perfect-electric-conductor (PEC) wire antenna, discretise it by the method of moments (MoM) with piecewise-constant basis functions and point-matching (collocation) testing, and analyse the resulting linear system from a numerical-analysis standpoint. The central object is the Pocklington kernel, in which the scalar- and vector-potential contributions are combined *before* evaluation so that the near-field $1/R^2$ and $1/R^3$ singularities cancel analytically; we show that this kernel is the unique formulation that is both pointwise valid and free of separate endpoint (charge) terms. We then confront the discretisation with the classical limits — the short-dipole radiation resistance, the half-wave dipole impedance, the resonant length, the far-field directivity, and the global power balance — and report what the solver reproduces to within a few percent and where the piecewise-constant basis degrades (very short and electrically long dipoles). As an application we design and simulate a dual-band VHF/UHF open-sleeve dipole for the 144–148 MHz (2 m) and 430–440 MHz (70 cm) amateur bands, computing feedpoint impedance, SWR, radiation pattern, directivity, gain, beamwidth, and front-to-back ratio on both bands. The design is well matched on the 2 m band (SWR $\lesssim 2.2$) but poorly matched on the 70 cm band (SWR $\sim$ 6–9); we show that this mismatch is set by the driven element, which is 1.36λ long on the upper band, and that the weakly-coupled parasitic cannot remove it — a concrete illustration of why "50 Ω on both bands" is a tuned-build claim, not a property of the geometry.

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1. Introduction

The VHF (30–300 MHz) and UHF (300 MHz–3 GHz) portions of the radio spectrum are the working bands of a large fraction of terrestrial wireless communication, and of the amateur-radio 2 m (144–148 MHz) and 70 cm (430–440 MHz) allocations. At these frequencies the electrical size of a practical wire antenna is of order a wavelength: a half-wave dipole on 2 m is about 1 m long, on 70 cm about 35 cm. This is the regime in which the *thin-wire integral-equation* description is both natural and accurate — the wire radius a is a few millimetres, three orders of magnitude below the wavelength, yet the wire is long enough that the current distribution is not trivial.

Analytical antenna theory handles the short dipole ($L \ll \lambda$) and, with more effort, the half-wave dipole ($L \approx \lambda/2$) in closed form. But a dual-band design — one structure that must behave acceptably at two frequencies a factor of three apart — has no closed-form answer: the current on each element is set by the coupled integral equations on *all* elements, and the feedpoint impedance on the upper band is controlled by the lower-band element seen at an electrical length of 1.36λ. This is a problem whose natural formulation is a (singular) integral equation, and whose natural solution is a numerical one. The method of moments is the standard discretisation.

The purpose of this essay is twofold. First, to derive the thin-wire EFIE and its MoM discretisation with the rigour that a functional/numerical-analysis reader expects: what the

integral operator is, what the basis and testing discretisation do to it, where the singularities live, what the quadrature must resolve, and what the convergence and conditioning look like. Second, to use the resulting solver as a *measuring instrument* on a real design problem, and to report — with the numbers the code actually produces — what is robust, what is a basis artefact, and what is physics.

We work throughout with the engineering time convention $e^{+j\omega t}$ and the free-space Green's function $G(R) = e^{-jkR}/(4\pi R)$, $k = \omega/c$. All numerical results in §5 are produced by the accompanying NumPy-only solver `antenna_mom.py` (Appendix A); the validation battery of §4 is `validate_fixed.py`.

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2. The thin-wire electric-field integral equation

2.1 From the potentials to the tangential field

Let the source be a z-directed surface current $I(z')$ on a straight PEC filament of radius a at transverse position (x_0, y_0), with time dependence $e^{+j\omega t}$. In the Lorenz gauge, $\nabla\cdot\mathbf{A} = -j\omega\mu_0\varepsilon_0\,\Phi$, the fields are expressed through the vector potential $\mathbf{A}$ and scalar potential Φ:

$$\mathbf{E} = -j\omega\mathbf{A} - \nabla\Phi, \qquad \mathbf{A}(\mathbf{r}) = \mu_0 \int \mathbf{J}(\mathbf{r}')\, G(\mathbf{r},\mathbf{r}')\, d^3r', \qquad G(R) = \frac{e^{-jkR}}{4\pi R}.$$

For a z-directed filament the only non-zero potential component is A_z, and $\nabla\cdot\mathbf{A} = \partial A_z/\partial z$, so

$$A_z(z) = \mu_0 \int I(z')\, G(z,z')\, dz', \qquad \Phi = -\frac{1}{j\omega\mu_0\varepsilon_0}\frac{\partial A_z}{\partial z}.$$

The tangential (z) scattered field on the wire is therefore

$$E^s_z(z) = -j\omega A_z(z) - \frac{\partial\Phi}{\partial z} = -j\omega\left[A_z(z) + \frac{1}{k^2}\frac{\partial^2 A_z}{\partial z^2}\right],$$

where we used $k^2 = \omega^2\mu_0\varepsilon_0$. Substituting the potential integral,

$$\boxed{E^s_z(z) = -j\omega\mu_0 \int I(z')\left[G(z,z') + \frac{1}{k^2}\frac{\partial^2 G(z,z')}{\partial z^2}\right] dz'} \tag{1}$$

with z the *observation* coordinate. The perfect-conductor boundary condition, $E^i_z(z) + E^s_z(z) = 0$ on the wire surface, gives the EFIE

$$j\omega\mu_0 \int I(z')\left[G + \frac{1}{k^2}\frac{\partial^2 G}{\partial z^2}\right] dz' = E^i_z(z). \tag{2}$$

Equation (2) is the starting point. It is a Fredholm integral equation of the first kind for the unknown current $I(z')$, with a kernel that is *weakly singular but not integrable in the naive form*: the second-derivative term $\partial^2 G/\partial z^2$ carries $1/R^3$, $1/R^4$, $1/R^5$ pieces that blow up as the observation point approaches the source ($R \to \rho$, the transverse separation).

2.2 The Pocklington kernel

The resolution, due to Pocklington [1], is to combine the scalar and vector potential contributions *before* evaluating the derivative, so that the near-field blow-ups cancel. Writing $u = z - z'$, ρ the transverse separation (equal to a for the self term, to the wire spacing for a mutual term), and $R = \sqrt{\rho^2 + u^2}$, one finds by direct differentiation

$$G + \frac{1}{k^2}\frac{\partial^2 G}{\partial z^2} = \frac{e^{-jkR}}{4\pi}\left[\frac{\rho^2}{R^3} + j\frac{2u^2 - \rho^2}{kR^4} + \frac{2u^2 - \rho^2}{k^2 R^5}\right],$$

and hence

$$-j\omega\mu_0\left[G + \frac{1}{k^2}\frac{\partial^2 G}{\partial z^2}\right] = \frac{j\omega\mu_0}{4\pi}\,K(z, z'),$$

where the **Pocklington kernel** is

$$\boxed{K = e^{-jkR}\left[\frac{u^2}{R^3} - \frac{2j\,u^2}{kR^4} - \frac{2\,u^2}{k^2R^5} + \frac{j\,\rho^2}{kR^4} + \frac{\rho^2}{k^2R^5} - \frac{1}{R}\right].} \tag{3}$$

Two structural facts about (3) matter for the numerics.

No separate endpoint term. The scalar-potential (charge) contribution is already *inside* the kernel integral. This is not an accident: it is the fundamental theorem of calculus. The charge density is $\rho_c(z') = (j/\omega)\,dI/dz'$ (continuity equation), and its potential contribution to E_z^s is $-\partial\Phi_c/\partial z$, which integrates by parts to a boundary term $g(z_1) - g(z_2) = -\int (dg/dz')dz'$ that is absorbed into the G term. Adding a *separate* endpoint/charge term therefore double-counts the scalar potential and corrupts the reactance. This is the first pitfall in constructing a thin-wire MoM code.

A sharp, narrow self-peak. As $R \to \rho$ (observation point on the surface directly above a source point, $u \to 0$), the kernel is dominated by the ρ^2/R^3 and $\rho^2/(k^2R^5)$ pieces, which peak with a width of order $\rho = a$. The self-term ($m = n$) matrix element therefore integrates a feature of width a — for $a = 3$ mm and a segment of length $\Delta\ell \sim 1$–5 mm the peak is comparable to or broader than the segment. This is the second, and numerically more serious, pitfall: a single coarse quadrature under-integrates the diagonal peak and collapses the impedance toward zero. §4 treats the quadrature this demands.

2.3 The EFIE as an operator

For the functional-analytic reader it is useful to recast (2) as an operator equation. Let $\mathcal{K}$ be the integral operator

$$(\mathcal{K}I)(z) = \frac{j\omega\mu_0}{4\pi}\int_{-L/2}^{L/2} K(z, z')\,I(z')\,dz',$$

acting on the space of (square-integrable) currents on the wire. The EFIE is

$$\mathcal{K}\,I = E_z^i. \tag{4}$$

$\mathcal{K}$ is a compact (Hilbert-Schmidt, modulo the weak $1/R$ singularity which is integrable) operator on L^2, so (4) is a regularised first-kind equation; the MoM replaces it by a finite-rank approximation. The choice of basis (the range of the finite-rank operator) and of testing (collocation vs. Galerkin) determines the discretisation error and, as we argue in §3.4, the *validity* of the kernel pairing.

3. Method-of-moments discretisation

3.1 Basis and expansion

We expand the current in **piecewise-constant** (PW-const) basis functions. The wire $[-L/2, L/2]$ is split into N segments of length $\Delta\ell = L/N$ with nodes $z_n = -L/2 + n\,\Delta\ell$, and

$$I(z) = \sum_{n=0}^{N-1} I_n\,\chi_n(z), \qquad \chi_n(z) = \begin{cases} 1, & z \in [z_n, z_{n+1}) \\ 0, & \text{otherwise.} \end{cases}$$

The coefficients I_n are the unknowns. The PW-const basis is the simplest choice and, with $\Delta\ell \ll \lambda$, is spectrally adequate for the half-wave regime that matters here. (The piecewise-*sinusoidal* basis, $f_n(z) = \sin(k(z - z_n))$, is a natural refinement — it enforces a smoother, approximately-continuous current and is more accurate in the short-dipole limit — but it changes the testing requirement, as discussed in §3.4.)

3.2 Collocation (point matching)

We enforce the EFIE **at a point**: at the centre z_m of each segment m, with the test point offset one radius a from the axis (on the wire surface). Substituting the PW-const expansion,

$$\sum_{n=0}^{N-1} I_n \underbrace{\frac{j\omega\mu_0}{4\pi} \int_{z_n}^{z_{n+1}} K(z_m, z')\,dz'}_{A_{mn}} = E^i_z(z_m),$$

gives the linear system

$$\mathbf{A}\,\mathbf{I} = \mathbf{b}, \qquad A_{mn} = \frac{j\omega\mu_0}{4\pi} \int_{z_n}^{z_{n+1}} K(z_m, z')\,dz', \tag{5}$$

where the right-hand side $\mathbf{b}$ is the incident field sampled at the test points. For a delta-gap feed of voltage V_0 at the centre, $E^i_z = V_0\,\delta(z)$, so $b_{\text{feed}} = V_0$ (dispersed as $V_0/\Delta\ell$ over the feed segment in the point-matching form) and $b_m = 0$ elsewhere.

3.3 The impedance matrix and the input impedance

The matrix $\mathbf{A}$ in (5) is the MoM impedance matrix. It is dense, complex, and non-symmetric in general (the kernel $K(z, z')$ is not symmetric in its arguments once the observation point is offset from the source axis). Its construction costs $O(N^2)$ kernel integrals, each a one-dimensional quadrature; the solve is $O(N^3)$ (or $O(N^2)$ per frequency for a fixed geometry with a factorised matrix).

The **input impedance** follows from the delta-gap model. With the PEC condition $E^s = -E^i$, the solved current on the feed segment I_{feed} relates to the applied voltage by

$$\boxed{Z_{\text{in}} = -\frac{V_0}{I_{\text{feed}}}} \tag{6}$$

The minus sign is a consequence of $E^s = -E^i$ and is a common source of sign errors: using $+V_0$ flips the sign of the radiation resistance, and a *negative* R_{rad} is the obvious red flag that the excitation sign is wrong.

For a **multi-wire** structure (a driven element plus open-circuited parasitics) the EFIE is enforced on *all* segments. The source is non-zero only at the driven element's feed segment; the parasitic segments have $E^i = 0$ (PEC, open circuit), so their currents are *induced* by the driven element. The feedpoint impedance of the driven element is Z_{in} from the full-system solve (6). For a true two-port (both elements fed) the open-circuit Z-parameters follow from the unit-current method: drive port p with $I_{\text{feed},p} = 1$ (replacing that row of $\mathbf{A}$ by the linear constraint $I_{\text{feed},p} = 1$), and read off $Z_{qp} = V_q / I_{\text{feed},p}$. A loaded two-port then reduces to $Z'_{11} = Z_{11} - Z_{12}Z_{21}/(Z_{22} + Z_L)$.

3.4 Why collocation, not Galerkin, with this kernel

This is the subtle point that the earlier (buggy) version of the solver got wrong, and it is worth stating precisely because it is a general feature of integral-equation discretisations.

The Pocklington kernel (3) is derived so that the EFIE (2) holds **pointwise** in the observation coordinate z. It is therefore the correct kernel for *collocation* (test at a point). If instead one multiplies (2) by a test function $f_m(z)$ and integrates over z (Galerkin), the term $(1/k^2)\,\partial^2 G/\partial z^2$ must be integrated by parts:

$$\int f_m(z)\,\frac{\partial^2 G}{\partial z^2}\,dz = \Big[f_m \frac{\partial G}{\partial z} - f'_m\,G\Big]_{\text{endpoints}} + \int f''_m(z)\,G\,dz.$$

Two things happen: (i) the second z-derivative on G is replaced by $f''_m(z)G$ (a derivative on the *test function*, not on the Green's function), and (ii) **endpoint terms** $[f_m\,\partial G/\partial z - f'_m G]$ appear. Feeding the un-integrated Pocklington kernel directly into a double (Galerkin) integral — $A_{mn} = \iint f_m K f_n\,dz\,dz'$ — skips *both* steps. It is not a discretisation of (2); it is a different, wrong operator.

This is exactly what the first version of `antenna_mom.py` did: a piecewise-sinusoidal basis with a Galerkin (double-integral) assembly of the un-integrated kernel. For a half-wave dipole it returned $Z \approx 0 - j1500\ \Omega$ instead of $\approx 73 + j42\ \Omega$, and the error did not converge with N (it is a formulation error, not a discretisation error, so mesh refinement cannot fix it). Replacing the Galerkin assembly by collocation with the Pocklington kernel (5) — the change made in the corrected solver — restores agreement with the classical values to $< 1\%$ in the half-wave regime (§4).

Lesson. The kernel and the testing procedure are a *pair*. The Pocklington kernel (3) is the collocation kernel. A Galerkin discretisation requires the integrated-by-parts kernel (with $f''_m G$ and endpoint terms), which is a different object. Mixing them is a category error that no amount of quadrature refinement can repair.

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4. Numerical analysis and validation

4.1 The quadrature: two-level Gauss–Legendre

The kernel (3) is smooth and bounded for $u \neq 0$ but peaks sharply at $u = 0$ on the **self** segment, with a width of order a. The off-diagonal terms ($m \neq n$) are smooth over the source segment. The correct response is a *two-level* quadrature:

- **Diagonal (self, $m = n$):** fine Gauss–Legendre, $n_q^{\text{self}} \approx 300$–$400$ nodes, to resolve the width-a peak.
- **Off-diagonal ($m \neq n$):** coarse Gauss–Legendre, $n_q^{\text{off}} \approx 24$ nodes, since the integrand is smooth.

A single coarse quadrature (say $n_q = 24$ everywhere) under-integrates the diagonal peak. In the buggy version this collapsed Z_{in} toward zero; with the corrected two-level scheme the half-wave anchors are reproduced to $< 1\%$ (§4.2). The cost of the fine diagonal is modest: only N of the N^2 elements use n_q^{self}.

4.2 Validation against the classical limits

The solver is validated against a battery of classical results before any design number is trusted. All results below are produced by `validate_fixed.py` at $f = 144$ MHz ($\lambda \approx 2.08$ m) unless stated.

(a) Half-wave dipole anchors. A centre-fed dipole of length 0.47λ and radius 3 mm should give $Z_{\text{in}} \approx 70 - j10\ \Omega$; one of 0.50λ and radius 1 mm should give $\approx 81 + j27\ \Omega$. The solver returns

Geometry	Z_{in} **(solver)**	**Reference**	**SWR @ 50** Ω
0.47λ, $a = 3$ mm, $N = 201$	$70.27 - j10.20\ \Omega$	$\approx 70 - j10$	1.46
0.50λ, $a = 1$ mm, $N = 201$	$81.39 + j27.52\ \Omega$	$\approx 81 + j27$	1.90

Agreement is to well under 1 % in both real and imaginary parts.

(b) Convergence in N. For the 0.47λ dipole:

N	Z_{in}	$\lvert\Delta Z\rvert$ **vs** N=401
101	$68.28 - j26.67$	2.34×10^{1}
201	$70.27 - j10.20$	6.80

301	$71.34 - j4.00$	5.32×10^{-1}
401	$71.56 - j3.52$	(ref)

The sequence converges monotonically; by $N = 301$ the change on further refinement is $< 0.6\ \Omega$. This is the normal PW-const convergence behaviour in the half-wave regime.

(c) Resonant length. Scanning L for the zero of the reactance gives $L/\lambda \approx 0.482$, against the classical 0.473 — a 2% length error, consistent with the finite radius ($a = 3$ mm shortens the resonant length).

(d) Power balance. For a unit feed current the input power $P_{\text{in}} = \frac{1}{2}\,\text{Re}(Z_{\text{in}})$ must equal the radiated power P_{rad} computed from the far-field Poynting integral (up to conductor loss). The solver gives $P_{\text{in}} = 35.1365$ W, $P_{\text{rad}} = 35.1372$ W, ratio 1.00002. This near-exact balance is a strong self-consistency check: it confirms that the impedance extraction (6) and the far-field normalisation are mutually consistent. It holds at all lengths tested, including the electrically-long dipoles of §4.3.

(e) Directivity and pattern. The 0.47λ dipole gives $D = 1.634$ (2.13 dBi) against the classical 2.15 dBi, an E-plane 3-dB beamwidth of 79° against $\approx 78^\circ$, and a radiating efficiency of 99.88% (copper surface resistance). All within a few percent of the textbook values.

4.3 Where the piecewise-constant basis degrades

Two limits expose the limitations of the PW-const basis. These are *basis* limitations, not kernel bugs — the power balance remains 1.0000 in both cases, so the solver is self-consistent; the issue is that the PW-const current is a poor representation of the true current in these limits.

(a) The short-dipole limit ($L \ll \lambda$). The radiation resistance of a short centre-fed dipole with (approximately) sinusoidal, feed-normalised current is $R_{\text{rad}} = 20\pi^2(L/\lambda)^2$ (the factor $80\pi^2$ belongs to the *uniform*-current Hertzian idealisation and must not be used as the comparison). The solver under-predicts R_{rad}, and the error shrinks as the dipole lengthens:

L/λ	R_{rad} **(solver)**	$20\pi^2(L/\lambda)^2$	**ratio**
0.02	0.0000	0.0790	0.00
0.05	0.1246	0.4935	0.25
0.10	1.4286	1.9739	0.72

The cause is visible in the current profile: for $L = 0.02\lambda$ the solved current is a narrow spike at the feed (peaking at 0.89 of its max and falling to ~ 0 within a segment or two) rather than a triangle spanning the wire. The PW-const staircase does not spread the

current over the short wire, so it radiates less than the analytical triangle. A piecewise-sinusoidal basis (which enforces a smoother, end-vanishing current) removes most of this error; we do not need it for the half-wave design of §5.

(b) The electrically-long limit ($L \gtrsim 0.7\lambda$). For a 0.75λ dipole the solved *current distribution* converges and has the correct "W" shape (a dip at the feed and two peaks, the standing-wave pattern of a 0.375λ arm), but the radiation resistance does **not** converge in N:

N	Z_{in}
101	$603 + j571\ \Omega$
201	$624 + j564\ \Omega$
401	$837 + j540\ \Omega$
601	$1049 + j391\ \Omega$

The real part climbs without settling as the mesh is refined. The power balance is still 1.0000 (the solver is self-consistent), so this is not a bookkeeping error; it is a genuine failure of the PW-const collocation discretisation to converge for electrically-long, weakly-ill-conditioned currents. This matters for §5 because, on the 70 cm band, the 2 m driven element is 1.36λ long and sits squarely in this regime. We treat the 70 cm feedpoint numbers as *indicative* (correct to order of magnitude and in sign) rather than converged, and we say so explicitly where they are used.

4.4 Summary of the validated regime

The corrected solver is reliable — to well under a few percent — for centre-fed dipoles and dipole-like elements in the range $0.2 \lesssim L/\lambda \lesssim 0.6$, i.e. the half-wave regime. This is exactly the regime of the dual-band design in §5 *on each element's own band*. The 70 cm feedpoint, which depends on the 2 m element at 1.36λ, is outside the validated regime and is reported as indicative.

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5. Applied design: a dual-band VHF/UHF open-sleeve dipole

5.1 The design problem

We design a single-structure antenna for the amateur 2 m band (144–148 MHz) and 70 cm band (430–440 MHz). A single straight dipole cannot be resonant at both bands — the frequencies are a factor of three apart, so a half-wave dipole would need to be simultaneously ≈ 1 m and ≈ 0.35 m long. The classic resolution is the **open-sleeve dipole**: a centre-fed 2 m driven dipole (resonant near 145 MHz) with a 70 cm parasitic element (resonant near 435 MHz) offset to one side. The parasitic is open-circuited; its current is induced. On the 2 m band the driven element radiates and the (electrically short) parasitic is a weak perturbation; on the 70 cm band the parasitic is near-resonant and re-radiates, while the driven element is far from resonant.

5.2 Geometry

The baseline geometry follows the well-known DK7ZB construction:

- **2 m driven element:** $L_2 = 936$ mm, radius $a_2 = 3$ mm, at $x = 0$, centre-fed. ($L_2 = 0.452\lambda$ at 145 MHz.)
- **70 cm parasitic:** $L_7 = 316$ mm, radius $a_7 = 3$ mm, at $x = +307$ mm. ($L_7 = 0.458\lambda$ at 435 MHz.)
- Mesh: $N_2 = 201$ segments on the driven element, $N_7 = 61$ on the parasitic; two-level quadrature $n_q^{\text{self}} = 300$, $n_q^{\text{off}} = 24$.

The full two-wire system ($N_2 + N_7 = 262$ unknowns) is solved at each frequency; the feedpoint is on the driven element.

5.3 Feedpoint impedance and SWR

The solver (`dualband_design.py`) sweeps both bands. The baseline geometry gives:

2 m band (144–148 MHz).

f **(MHz)**	Re Z_{in}	Im Z_{in}	**SWR @ 50** Ω
144	60.1	-43.5	2.21
145	61.6	-38.3	2.02
146	63.1	-33.0	1.86
147	64.6	-27.7	1.72
148	66.1	-22.4	1.61

The 2 m band is moderately matched (SWR 1.6–2.2) and *improves* with frequency: the driven element is 0.452λ (slightly short of the 0.473λ resonance), so it is capacitive, and the reactance falls toward zero as the band centre is approached from below. This is the well-validated regime (§4.4), so these numbers are reliable.

70 cm band (430–440 MHz).

f **(MHz)**	Re Z_{in}	Im Z_{in}	**SWR @ 50** Ω
430	66.7	-155.2	9.19

435	67.8	-137.7	7.55
440	69.9	-120.1	6.07

The 70 cm band is poorly matched (SWR 6–9), dominated by a large *capacitive* reactance. This is the regime flagged in §4.3(b): the 2 m driven element is $936\,\text{mm} = 1.36\lambda$ at 435 MHz, far from any resonance, and its impedance — not the parasitic's — sets the feedpoint.

5.4 Why the parasitic cannot fix the 70 cm mismatch

A natural hope is that tuning the 70 cm parasitic (its length L_7 and offset x_7) can cancel the capacitive reactance. It cannot, and the solver demonstrates why. Sweeping $L_7 \in \{280, \ldots, 360\}$ mm and $x_7 \in \{240, \ldots, 400\}$ mm (25 geometries) at 435 MHz, the SWR moves only between 7.55 and 7.62 — the feedpoint is essentially *insensitive* to the parasitic:

$$Z_{\text{in}}(435\,\text{MHz}) \approx 67.8 - j137.6\ \Omega \quad \text{for every geometry tested.}$$

The reason is the two-port structure. Computing the open-circuit Z-parameters of the two-port (driven element = port 1, parasitic = port 2) at 435 MHz gives

Parameter	**Value**	**Meaning**
Z_{11}	$67.8 - j137.6\ \Omega$	driven-element self-impedance (parasitic open)
Z_{12}	$-0.25 + j1.99\ \Omega$, $\lvert Z_{12}\rvert = 2.0\ \Omega$	mutual impedance
Z_{22}	$73.3 - j10.9\ \Omega$	parasitic self-impedance (near the half-wave $73\ \Omega$)

The coupling coefficient $k = |Z_{12}|/\sqrt{|Z_{11}Z_{22}|} \approx 0.02$ is very small: the parasitic is weakly coupled to the driven element. The parasitic loads the driven element by the correction

$$Z'_{\text{in}} = Z_{11} - \frac{Z_{12}^2}{Z_{22}} \approx 67.8 - j137.6\ \Omega + (0.01 + j0.02)\ \Omega,$$

which moves the feedpoint by only $\approx 0.02\ \Omega$. The parasitic therefore cannot cancel the $-j138\ \Omega$ capacitive reactance of the driven element at 1.36λ; it shapes the *radiation pattern* on the 70 cm band (it is near-resonant and re-radiates) but does not significantly change the *feedpoint impedance*. This is the precise sense in which "50 Ω on both bands" is a tuned-build claim (a matching network, a different topology, or a feed-point transformation) rather than a property of the open-sleeve geometry.

5.5 Radiation pattern, directivity, and gain

The far field is the coherent sum of Hertzian elements (§3.3, Appendix A). For the optimised geometry the pattern metrics are:

Quantity	**145 MHz (2 m)**	**435 MHz (70 cm)**
Directivity D	1.70 (2.30 dBi)	3.69 (5.67 dBi)
Gain G	1.69 (2.29 dBi)	3.68 (5.15 dBi)
Radiating efficiency	99.4%	99.8%
E-plane 3-dB beamwidth	86°	32°
H-plane 3-dB beamwidth	134°	58°
Front-to-back ratio	0.1 dB	1.6 dB

On the 2 m band the pattern is essentially that of a half-wave dipole: directivity 2.30 dBi (classical 2.15 dBi), an E-plane beamwidth of 86° (classical $\approx 78^\circ$), and a nearly-omnidirectional H-plane, as expected for a z-directed dipole.

On the 70 cm band the pattern is that of the electrically-long driven element (1.36λ): a higher directivity (5.67 dBi) and a narrow E-plane beam (32°), the multi-lobed pattern of a long dipole. The front-to-back ratio is small (1.6 dB) because the parasitic is weakly coupled (§5.4) and cannot create strong front-back asymmetry; the directionality comes from the long driven element, not from the parasitic. Radiating efficiency is high on both bands (99.4%, 99.8%) because copper loss in a 3 mm wire at VHF/UHF is small, so gain sits just below directivity.

5.6 Design assessment

The open-sleeve dipole is a good 2 m antenna (SWR $\lesssim 2.2$, near the classical half-wave directivity) and a functional but poorly-matched 70 cm antenna (SWR $\sim$ 6–9). For a 70 cm band that must drive a $50\ \Omega$ transmitter without a tuner, the feedpoint needs a matching network (e.g. a lambda/8 choke or a gamma match) or a different topology. The MoM analysis has the value of *quantifying* this before any metal is bent: it predicts the 2 m match, predicts the 70 cm mismatch, and shows — by the insensitivity of the feedpoint to the parasitic — exactly where the mismatch lives.

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6. Discussion

The exercise illustrates three points that are general to integral-equation antenna analysis.

Formulation errors do not converge. The Galerkin-with-unintegrated-kernel bug of §3.4 produced a Z_{in} that was wrong by a factor of ~ 20 and *did not improve with mesh refinement*, because it discretised the wrong operator. This is a caution for any MoM code: a result that does not converge with N is a signal of a formulation (or sign) error, not a discretisation error, and no quadrature refinement will fix it. The correct diagnostic is to check the *classical limits* (half-wave impedance, short-dipole R_{rad}, power balance) before trusting any design number.

Basis choice sets the validated regime. The PW-const basis is accurate in the half-wave regime ($0.2 \lesssim L/\lambda \lesssim 0.6$) and degrades in the short- and long-dipole limits (§4.3). The power balance staying at 1.0000 even where the impedance is wrong shows that self-consistency (Poynting balance) is necessary but not sufficient for accuracy — it checks the *normalisation*, not the *current distribution*.

A single number can hide the physics. The 70 cm SWR of ~ 7 looks like a "bad design," but the two-port analysis (§5.4) localises it: it is the driven element at 1.36λ, and the parasitic is too weakly coupled to matter at the feedpoint. The MoM result is not just a pass/fail SWR; it is a diagnostic of *where* the mismatch is, which is what a designer needs.

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7. Conclusion

We derived the thin-wire Pocklington EFIE, discretised it by MoM with piecewise-constant basis and collocation testing, and analysed the discretisation's singular structure, quadrature requirements, and convergence. The corrected solver reproduces the classical half-wave dipole impedance, resonant length, directivity, and power balance to well under a few percent, and its validated regime ($0.2 \lesssim L/\lambda \lesssim 0.6$) covers each element of a dual-band VHF/UHF open-sleeve dipole on its own band. Applied to a 144–432 MHz design, the solver predicts a well-matched 2 m band (SWR $\lesssim 2.2$) and a poorly-matched 70 cm band (SWR $\sim$ 6–9), and — through the two-port Z-parameters — shows that the 70 cm mismatch is set by the driven element at 1.36λ, not by the parasitic, and cannot be removed by tuning the parasitic. The method is a reliable measuring instrument for wire-antenna design in the half-wave regime, and its limitations (short- and long-dipole basis degradation) are clearly identifiable.

References

[1] H. Pocklington, "On the Calculation of Electrostatic and Electromagnetic Problems by Means of a System of Linear Equations," *Proc. Phys. Soc. London*, ser. A, vol. 36, pp. 537–552, 1924. (The original method-of-moments formulation for wire antennas; the kernel (3) is the Pocklington kernel.)

[2] R. F. Harrington, *Field Computation by Moment Methods*. New York: IEEE Press / W. H. Freeman, 1968. (The standard treatment of the MoM: basis and testing functions, the EFIE, and the thin-wire kernel.)

[3] C. A. Balanis, *Antenna Theory: Analysis and Design*, 4th ed. Hoboken, NJ: Wiley, 2016. (Classical dipole impedances, the short-dipole $R_{\text{rad}} = 20\pi^2(L/\lambda)^2$, and the half-wave dipole reference values used in §4.)

[Note on citation verification: references [1]–[3] are standard, well-established works. Specific page numbers are intentionally omitted; the results attributed to them (the Pocklington kernel, the MoM discretisation, the classical dipole impedances) are the canonical content of each text. Verify page-level detail against the physical editions before formal publication.]

Appendix A: The solver

`antenna_mom.py` (NumPy only) implements:

- `Wire(x0, y0, z1, z2, a, N)` — a straight PEC wire.
- `build_Z(wires, f, nq_off, nq_self)` — the MoM impedance matrix (5), two-level Gauss–Legendre quadrature, Pocklington kernel (3), no endpoint term.
- `solve(wires, f, feed_seg, V0)` — (Z_{in}, I) from the delta-gap model (6), $Z_{\mathrm{in}} = -V_0/I_{\mathrm{feed}}$.
- `z_parameters(wires, f, feed_segs)` — open-circuit n-port Z-parameters (unit-current method).
- `far_field`, `radiated_power`, `directivity`, `efficiency`, `swr`, `s11`, `pattern_metrics` — far-field and pattern quantities.

The validation battery `validate_fixed.py` runs the checks of §4; `dualband_design.py` runs the design of §5 and writes `dualband_results.json` and `dualband_sweeps.csv`.